

Metaphysical Foundations, Multi-Agent Swarm Invariants, and Singularity Quenching

System Architecture Invariant Manifest

September 29, 2026

1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta's Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past.
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites.
3. **The Indexical Real World:** The concrete layer where sovereign actions occur in real time.

2 The DSL Discretization Interface and Continuous Superposition

Continuous environments present an infinite, unsearchable behavior space. We address this challenge by introducing a LISP-like Domain-Specific Language (DSL) that functions as a structural discretization filter.

3 Multi-Agent Weather-Adaptive Swarm Invariants (Tree #12)

To capture the mathematical intuition of distributed control, we model a swarm of $N = 6$ vehicles navigating a continuous, turbulent space-time

weather manifold.

3.1 Agent Kinematics (ODE Layer)

Each drone $i \in \{1, \dots, N\}$ is governed by decoupled 2D double-integrator dynamics:

$$\dot{\mathbf{p}}_i(t) = \mathbf{v}_i(t), \quad \dot{\mathbf{v}}_i(t) = \mathbf{u}_i(t) \quad (1)$$

where $\mathbf{p}_i = [x_i, y_i]^T \in \mathbb{R}^2$ is the position vector, $\mathbf{v}_i \in \mathbb{R}^2$ is the velocity vector, and $\mathbf{u}_i \in \mathbb{R}^2$ is the acceleration control command vector.

3.2 Turbulent Field Input (PDE Layer)

The localized weather disturbance is governed by a 2D hyperbolic wave equation tracking a scalar intensity field $\phi(x, y, t)$ deformed by a non-linear Navier-Stokes advection velocity layer $\mathbf{v}_{\text{turb}}$ satisfying a Kolmogorov power law:

$$\frac{\partial^2 \phi}{\partial t^2} + \mathbf{v}_{\text{turb}} \cdot \nabla \left(\frac{\partial \phi}{\partial t} \right) = c^2 \nabla^2 \phi \quad (2)$$

3.3 The Multi-Objective Control Law

The total acceleration command vector $\mathbf{u}_i(t)$ resolves three distinct geometric and physical constraints:

$$\mathbf{u}_i = \mathbf{u}_i^{\text{formation}} + \mathbf{u}_i^{\text{weather}} + \mathbf{u}_i^{\text{collision}} \quad (3)$$

3.3.1 Term 1: Consensus Formation Keeping

Maintains the rigid geometric formation specified by structural neighbor offsets $\boldsymbol{\delta}_{ij}$:

$$\mathbf{u}_i^{\text{formation}} = -k_p \sum_{j=1}^N a_{ij} (\mathbf{p}_i - \mathbf{p}_j - \boldsymbol{\delta}_{ij}) - k_v \sum_{j=1}^N a_{ij} (\mathbf{v}_i - \mathbf{v}_j) \quad (4)$$

where $k_p, k_v > 0$ are fixed feedback gains, and $a_{ij} = 1$ models the static all-to-all perfect communication link.

3.3.2 Term 2: Wavelet Filtered Feedforward

Extracts localized, scale-invariant spatial gradients using a Discrete Wavelet Transform (DWT) operator to prune out high-frequency chattering noise:

$$\mathbf{u}_i^{\text{weather}} = -k_w \cdot \text{DWT}(\nabla \phi(\mathbf{p}_i, t)) \quad (5)$$

3.3.3 Term 3: Artificial Potential Field Collision Avoidance

Guarantees safety constraints by enforcing an unresolvable repulsion boundary if the inter-agent Euclidean distance $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|$ violates a critical threshold:

$$\mathbf{u}_i^{\text{collision}} = - \sum_{j \neq i} \nabla_{\mathbf{p}_i} V_{ij}(d_{ij}) \quad (6)$$

where the potential barrier function $V_{ij}(d_{ij})$ is defined over the influence range $d_{\text{influence}}$ and critical crash radius r_{safe} as:

$$V_{ij}(d_{ij}) = \begin{cases} \frac{1}{2}\eta \left(\frac{1}{d_{ij}-r_{\text{safe}}} - \frac{1}{d_{\text{influence}}-r_{\text{safe}}} \right)^2 & \text{if } d_{ij} < d_{\text{influence}} \\ 0 & \text{if } d_{ij} \geq d_{\text{influence}} \end{cases} \quad (7)$$

with scaling parameter $\eta > 0$.